NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

MIL-PRF-22/15A
21 May 1971
SUPERSEDING
MIL-R-22/15
7 January 1969

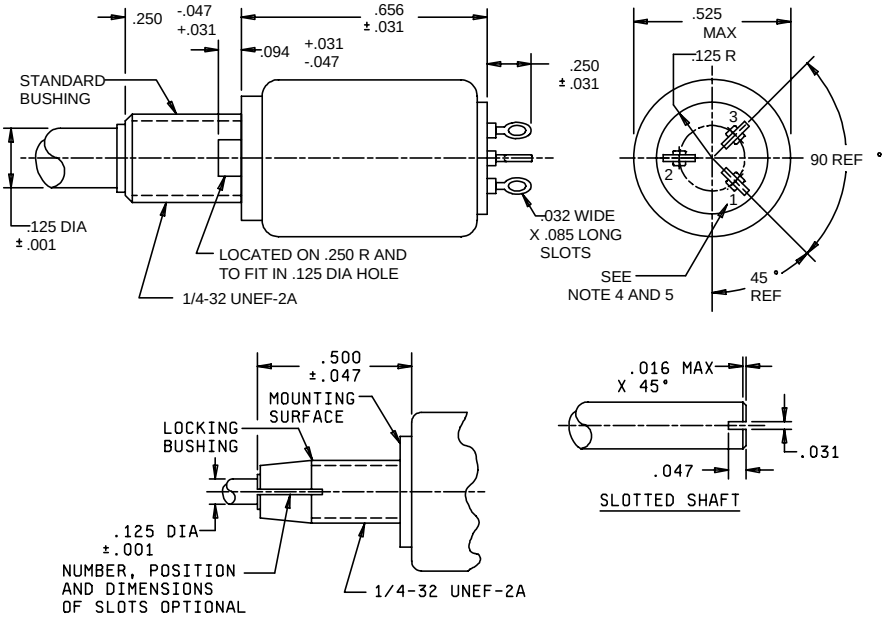
PERFORMANCE SPECIFICATION

RESISTOR, VARIABLE, (WIREWOUND, POWER TYPE)

(ENCLOSED), STYLE RP05

The complete requirements for procuring the resistors described herein shall consist of this document and the latest issue of the Specification MIL-R-22.

This specification is mandatory for use by all Departments and agencies of the Department of Defense.



INCHES	MM
.001	.03
.005	.05
.016	.13
.031	.79
.032	.81
.047	1.19
.085	2.16
.094	2.39
.125	3.18
.250	6.35
.500	12.70
.525	13.34
.656	16.66

NOTES:
1. All dimensions in inches.
2. Metric equivalents (to the nearest .01mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Unless otherwise specified, tolerance is + .005 (.13mm)
4. Terminal identification for reference only.
5. Design and position of terminals optional

Figure 1 - Outline and dimensions, Style RP05

(A) denotes changes

FSC 5905

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REQUIREMENTS:

Dimensions and configuration: See figure and table I.

Resistance value: See table I.

Resistance tolerance: ±10 percent.

Electrical off position: No electrical off position.

Shaft style and type of mounting: See table III.

Shaft length: See table IV.

Mechanical rotation: 300° ±5°.

Hardware: One 1/4-32 UNEF-2B corrosion-resistant, hexagonal mounting nut having a nominal thickness of 1/16 inch and measuring 5/16 inch across the hexagonal flats, and one corrosion-resistant, internal-tooth lockwasher having an outside nominal diameter of 0.402 inch, inside nominal diameter of 0.262 inch, and a nominal thickness of 0.013 inch shall be supplied. The locking nut for locking-bushing-type resistors shall be 5/32-inch thick, and shall measure 5/16 inch across the hexagonal flats; the thread size shall be 1/4-32 UNEF-2B.

Dielectric withstanding voltage:
At atmospheric pressure - Method 301 of MIL-STD-202. Test voltage 500 volts root mean square (rms).
At reduced pressure - Method 105 of MIL-STD-202. Test voltage - 250 volts.

DC resistance: Total and minimum. Method 303 of MIL-STD-202.

Total resistance within ±10 percent of the nominal total resistance specified in table I.

Minimum resistance shall not exceed 0.2 percent of the nominal total resistance value or 1.5 ohms, whichever is greater.

Torque: Operating - Not less than 0.25 ounce-inches, nor more than 3.0 ounce-inches. Stop - 5 pound-inches. Locking - Shall withstand 20 ounce-inches without movement.

Low-temperature exposure: Torque not in excess of 8 ounce-inches.

Power rating: 5.0 watts at 25° C. Mounted on a steel panel 4 inches square by 0.040 inches thick. Hot spot not in excess of 340° C.

Load life: Full wattage (see power rating, above) applied 1-1/2 hours on and 1/2 hour off for 1,000 hours. Change in total resistance not to exceed ±5 percent.

Vibration: Not applicable (see high frequency vibration).

Salt spray (corrosion): Method 101 of MIL-STD-202. Test condition B (48 hours).

Humidity (steady state): Method 103 of MIL-STD-202. Test condition B (96 hours). Change in total resistance shall not exceed ±10 percent.

Life (rotational): Method 206 of MIL-STD-202. Test condition A (500 cycles) for locking bushing types. Test condition C (5,000 cycles) for standard bushing types. Change in total resistance not to exceed ±5 percent.

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Terminal strength: No mechanical damage with 10-pound load applied each terminal for 1 minute. Change in total resistance shall not exceed ±1 percent.

(A) Shock: Method 213 of MIL-STD-202, Test condition G. Change in resistance shall not exceed ±10 percent due to contact arm movement and the total resistance change shall not exceed ±2 percent.

TABLE I. Style RP05.

Part number 1/	Nominal total resistance value (ohms)	Maximum current (not to be exceeded on any portion of the winding) (amperes)	Type designation 1/ (for reference only)
M22/15-0001---	10	.707	RP05---100KK
M22/15-0002---	15	.583	RP05---150KK
M22/15-0003---	25	.447	RP05---250KK
M22/15-0004---	35	.374	RP05---350KK
M22/15-0005---	50	.316	RP05---500KK
M22/15-0006---	75	.264	RP05---750KK
M22/15-0007---	100	.223	RP05---101KK
M22/15-0008---	150	.182	RP05---151KK
M22/15-0009---	200	.158	RP05---201KK
M22/15-0010---	250	.141	RP05---251KK
M22/15-0011---	350	.118	RP05---351KK
M22/15-0012---	500	.1	RP05---501KK
M22/15-0013---	750	.082	RP05---751KK
M22/15-0014---	1,000	.071	RP05---102KK
M22/15-0015---	1,500 2/	.056	RP05---152KK
M22/15-0016---	2,500 2/	.045	RP05---252KK
M22/15-0017---	3,500 2/	.037	RP05---352KK
M22/15-0018---	5,000 2/	.032	RP05---502KK

1/ The complete type designation and part number include symbols indicating electrical off position from table II, style of shaft and type of mounting from table III, and length of operating shaft from table IV (see example of part number).

2/ Values based on use of wire size smaller than .0025 inch but not less than .0014 inch.

TABLE II. Electrical off position.

Symbol	Electrical off position
1 - - - - - - - - - -	No electrical off position

TABLE III. Style of shaft and type of mounting.

Symbol	Style of shaft	1/8-inch diameter shaft	
		Standard bushing	Locking bushing
S - - - - -	Slotted	X	---
U - - - - -	Slotted	---	X

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TABLE IV. Length of operating shaft.

Standard length of shaft measured from mounting surface of resistor.		
Symbol	Slotted	
	1/8-inch diameter shaft	
	Standard bushing	Locking bushing
	Inches, ±3/64	Inches, ±3/64
B - - - - -	5/8	5/8
D - - - - -	---	7/8

Vibration, high frequency: Method 204 of MIL-STD-202. Test condition B (15G), 10 to 2, 000 Hz. Movement of contact arm not to result in a resistance change greater than ±10 percent. Total resistance change not to exceed ±2 percent. No mechanical damage.

Acceleration: Not applicable to this specification.

Part number: Consists of the basic number of this specification sheet, a dash number from table I, and electrical and mechanical characteristics taken from tables II, III, and IV.

EXAMPLE:

M22/15-	0001	1	S	B
Basic document number (indicating MIL-R-22/15)	Dash number from table I (indicating 10 ohms)	Electrical off position from table II (indicating off position not used)	Style of shaft and type of mounting from table III (indicating standard bushing, slotted shaft)	Length of shaft from table IV (indicating 5/8-inch shaft)

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army -
Navy - AS, OS
Air Force - 17, 80
DSA - ES

User activities:
Army - ME, AT, AV
Navy - CG
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5905-0755)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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